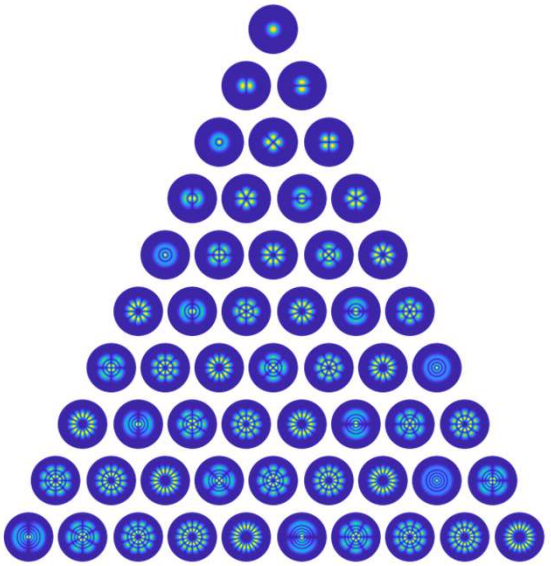
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

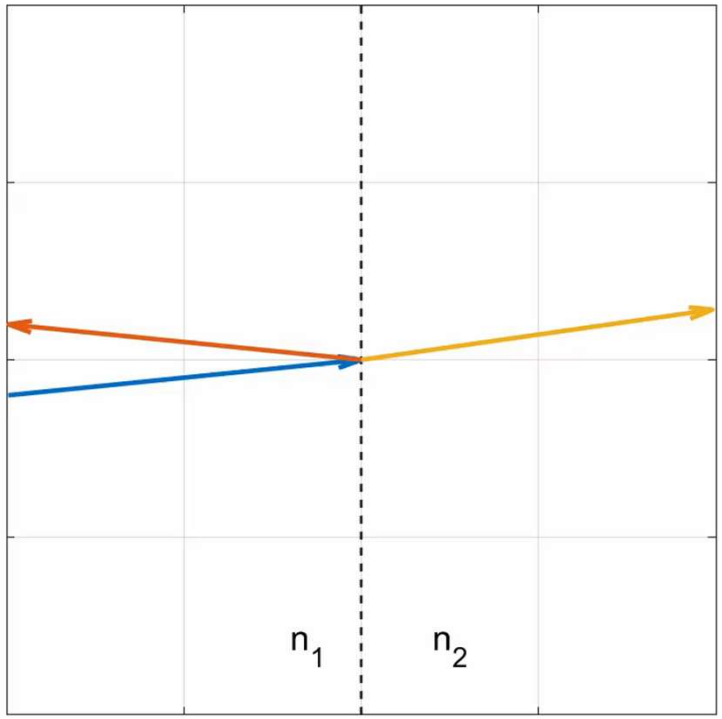
Lecture 4



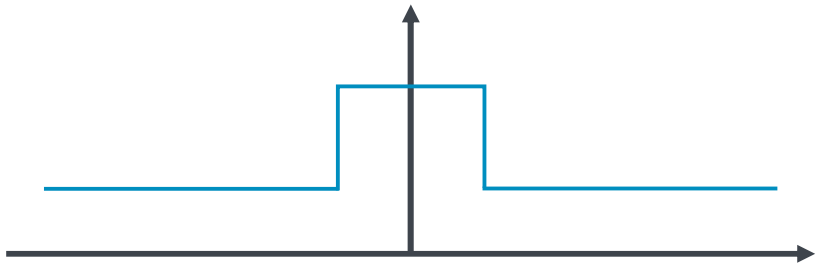
Prof. Dr.-Ing. Georg Rademacher

Snell's law



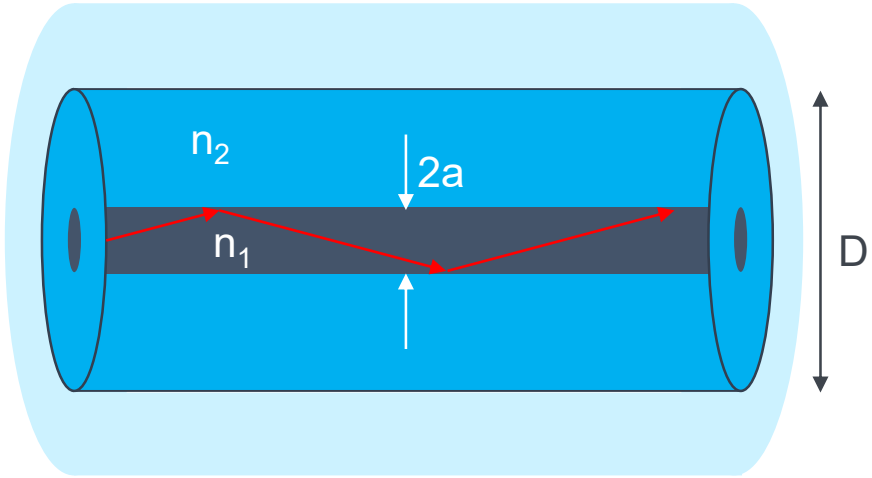


Step-index fiber

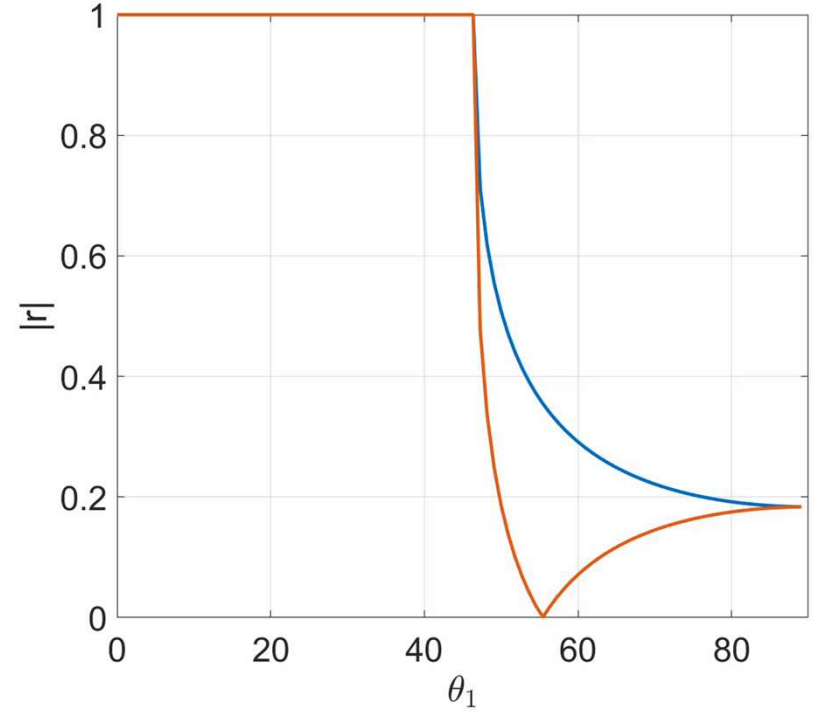


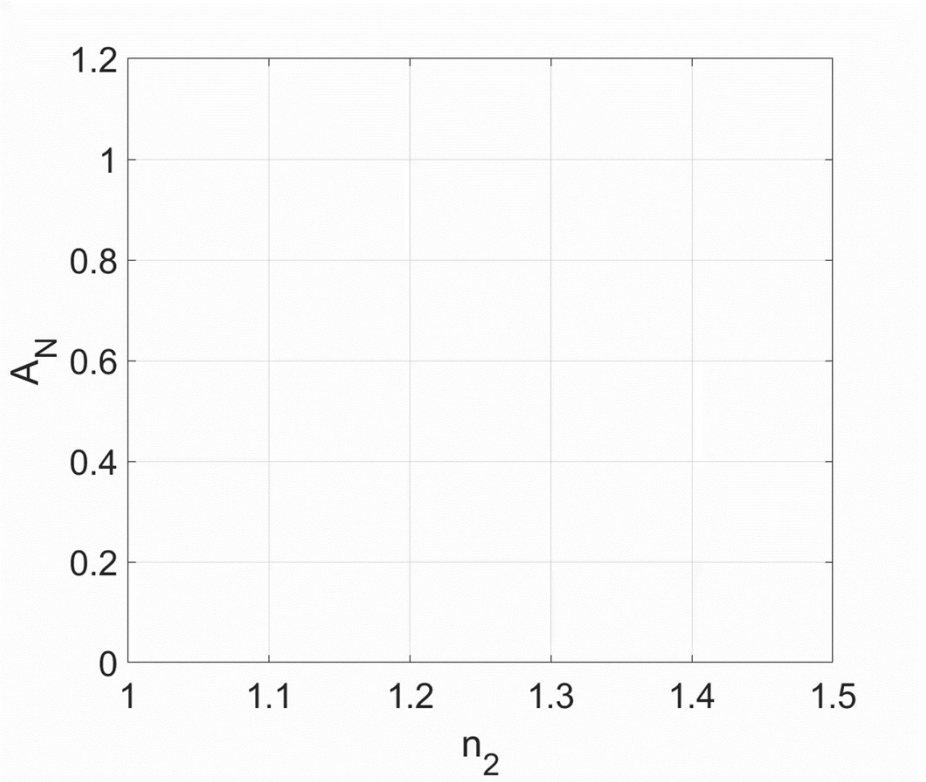
Step-index fiber

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Germany



Step-index fiber





Electromagnetc fields in step-index fiber

Electromagnetc fields in step-index fiber

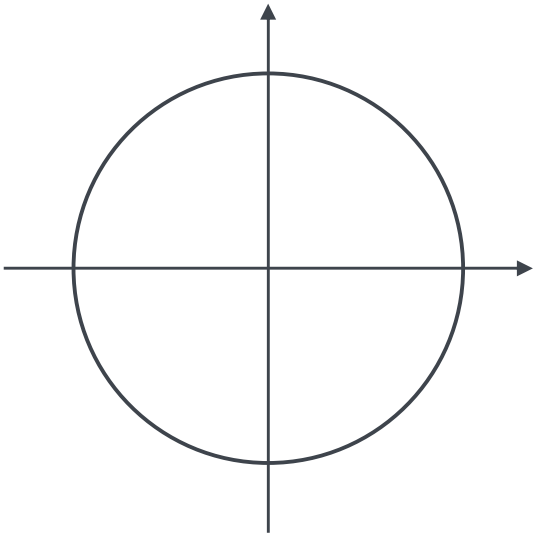
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Electromagnetc fields in step-index fiber

Electromagnetc fields in step-index fiber

Electromagnetc fields in step-index fiber

Boundary conditions in step-index fiber



Boundary conditions in step-index fiber

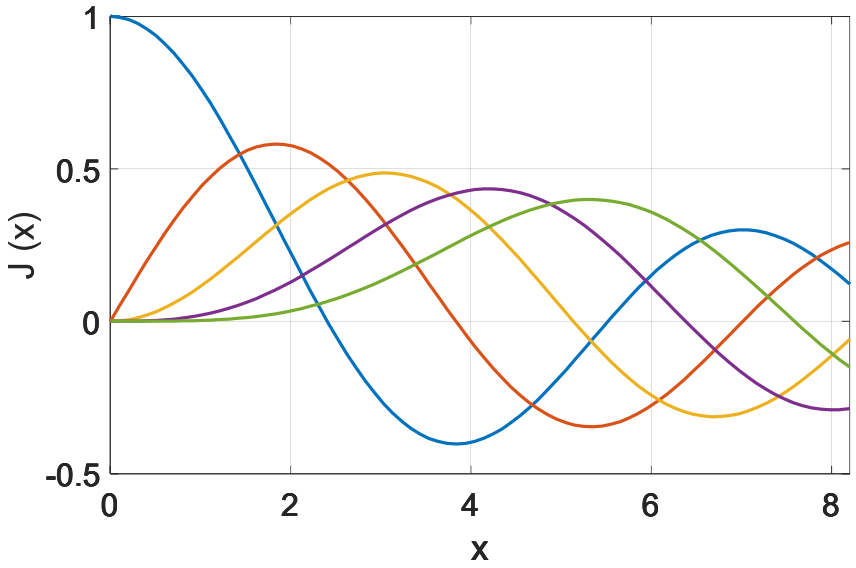
Linearly polarized (LP) waves

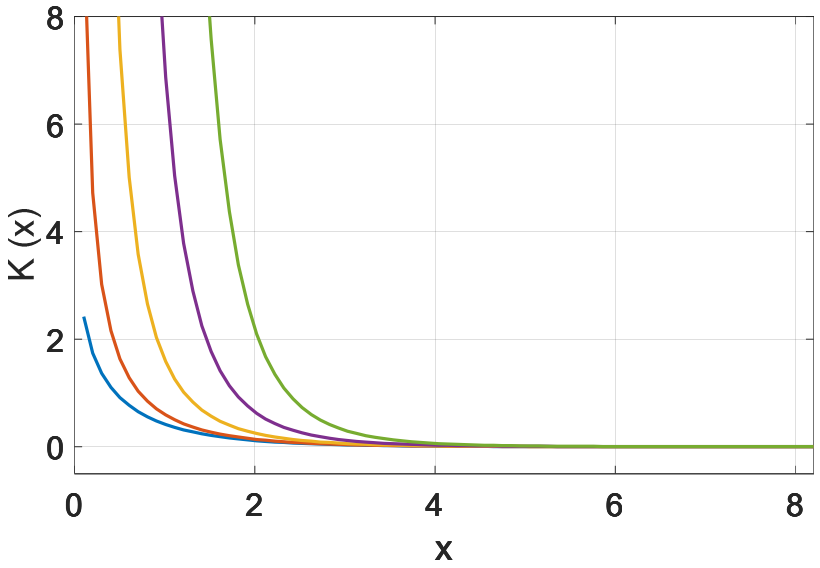
Linearly polarized (LP) waves

Linearly polarized (LP) waves

Besselfunctions

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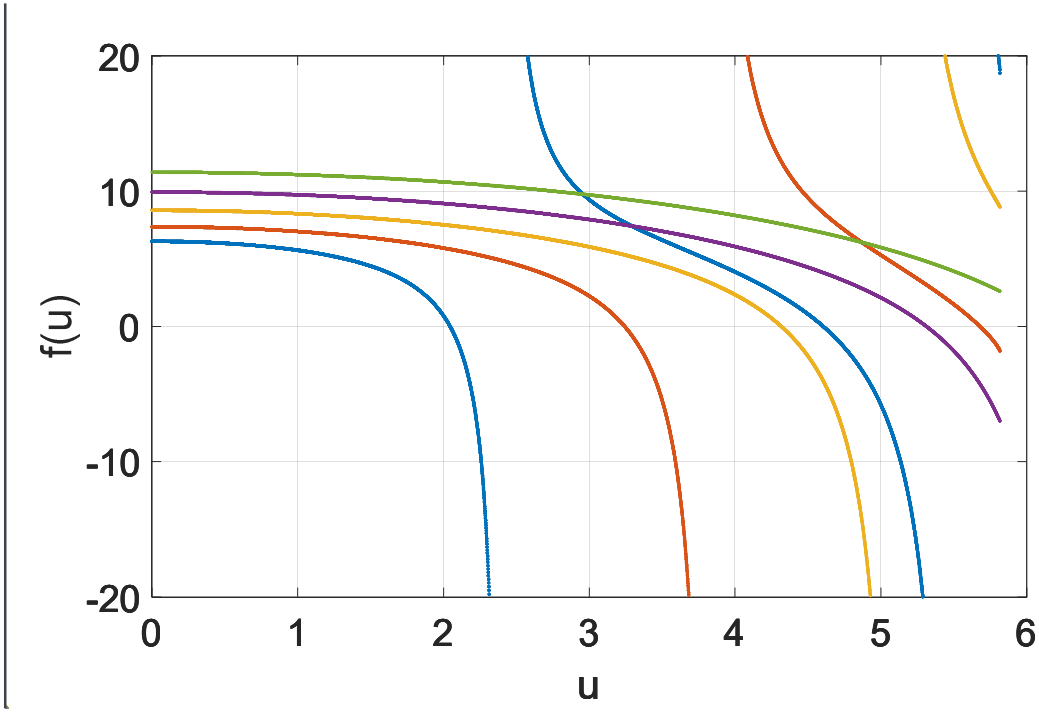




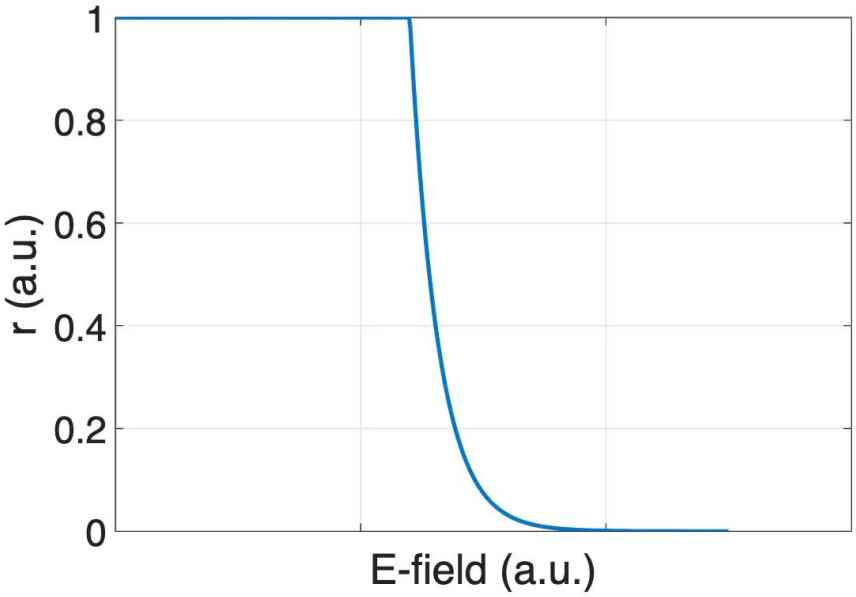
Linearly polarized (LP) waves

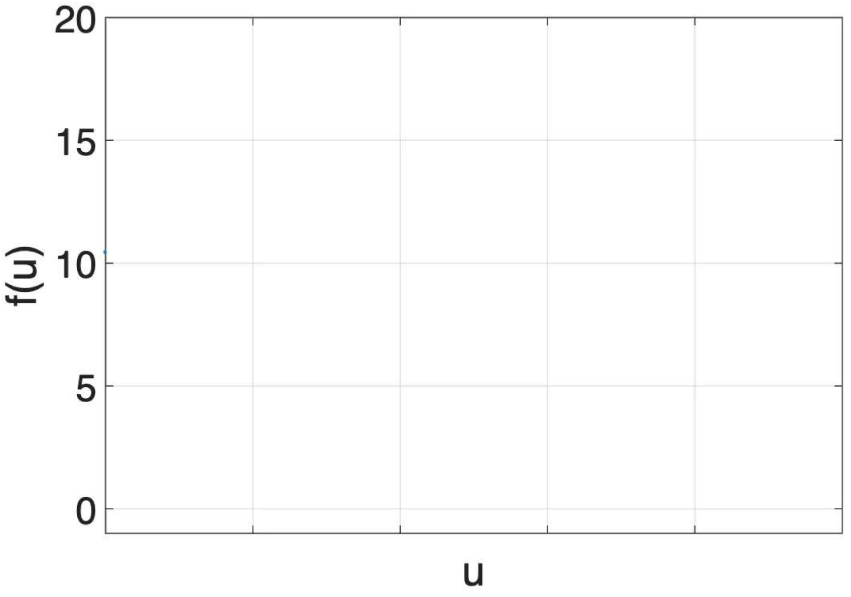
Linearly polarized (LP) waves

Linearly polarized (LP) waves



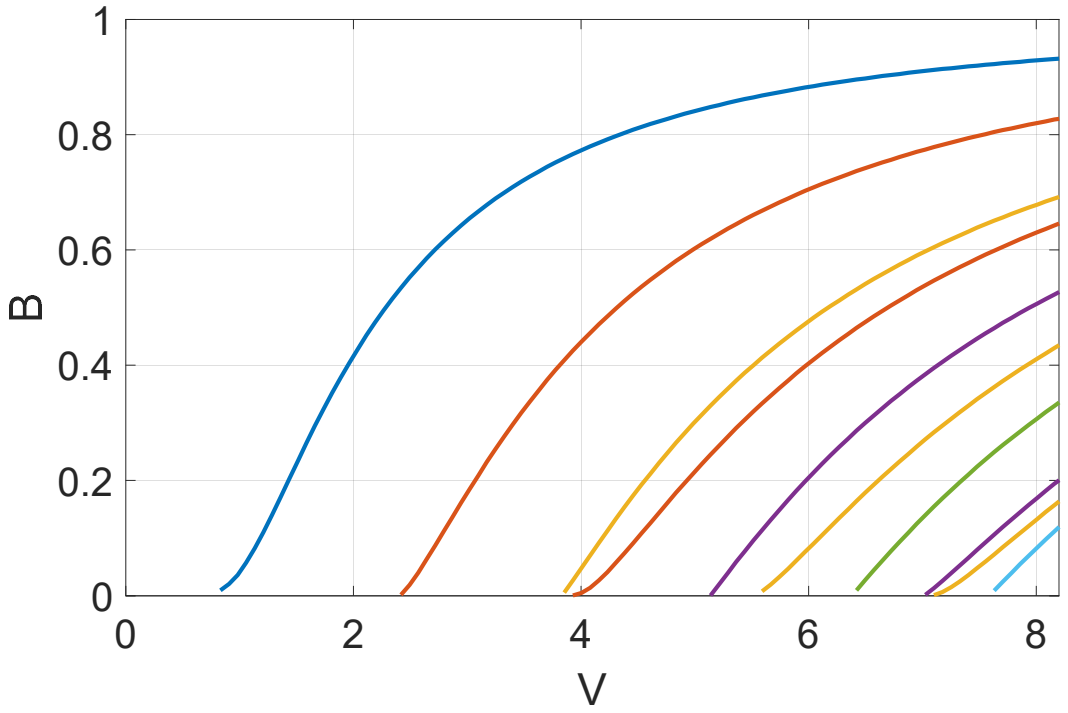
Linearly polarized (LP) waves





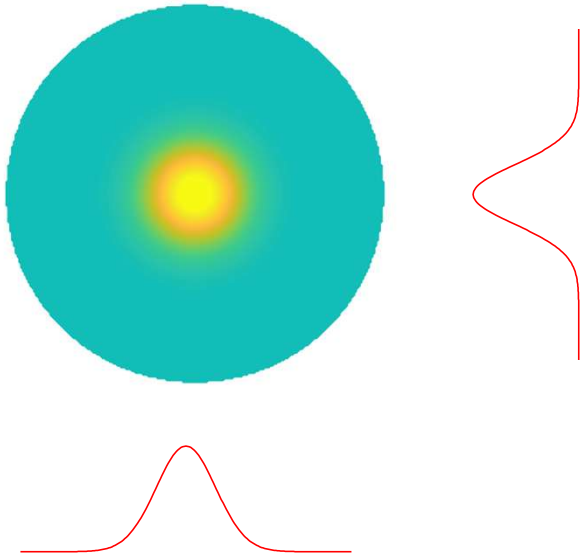
Linearly polarized (LP) waves

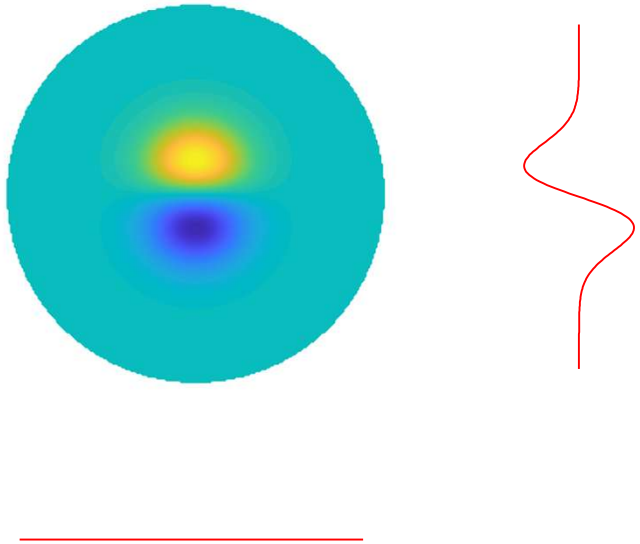
Linearly polarized (LP) waves



Mode Fields in Step Index Fibers

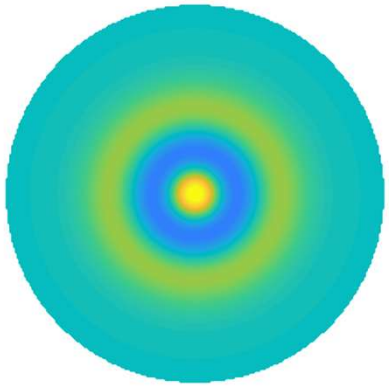
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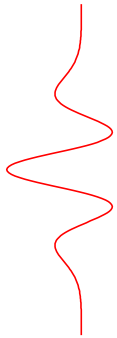


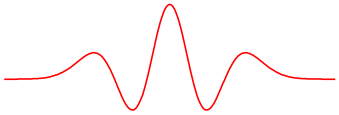


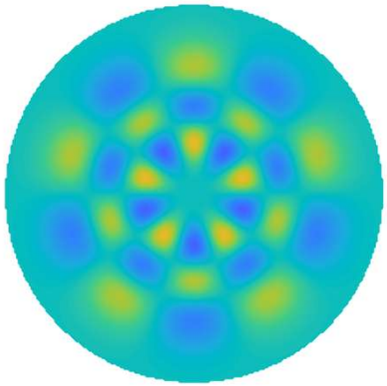
Mode Fields in Step Index Fibers

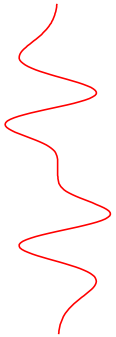
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Single-mode fiber

Single-mode fiber

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Institute of electrical and optical communications

Thank you!



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